# PROJECT AND TEAM INFORMATION

## Project Title

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| *Emergency Blood Donor Matching System* |

## Student / Team Information

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| *Team Name:*<br>*Team # DSCPP-III-2026-T092* | *LifeLink* |
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# PROPOSAL DESCRIPTION

## Motivation

*In my own village, I have seen cases where people have died due to unavailability of blood whenever it was required. During those moments, I did not have the requisite knowledge and skills which could enable me to provide assistance in any way. Such instances have informed me of the importance of having timely access to blood in emergencies. Today, as an IT student who has knowledge in programming and technology, I am sure that I have the ability to solve this problem through what I know.*
*The* ***Emergency Blood Donor Matching System*** *is our project aimed at helping individuals identify potential blood donors in emergencies. The project aims at matching patients or their relatives with individuals who are potential blood donors. This will be done using factors such as blood group among others. The main objective of this project is to ensure that there is reduction of time used in identifying potential donors in emergencies.*

## State of the Art / Current solution

*At this time, most people typically turn to local blood banks, hospitals, relatives and/or friends along with other online or offline social resources. (i.e. WhatsApp and Facebook) for finding potential blood donors. There are also several web-based platforms that allow one to look for the availability of blood in your area, find local blood banks, and also sign-up as a blood donor through sites like e-RaktKosh. Although social media has greatly assisted in rapidly disseminating information regarding an immediate need for blood donations, unfortunately, it does not have mechanisms by which patients may match up with potential donors who are nearby and available at the same time based upon their blood type, geographical location, etc. The goal of this project is to provide a platform where users can quickly post the immediate need for blood and utilize certain criteria to identify a suitable blood donor(s). In addition, our project will assist patients in locating available blood donors during emergencies but we are not attempting to displace hospitals or to take over the responsibility of managing official blood banks.*

## Project Goals and Milestones

*The primary goal of this project that we have developed through a system called Emergency Blood Donor Matching System is to provide an effective solution to find the closest blood donor compatible with patient's blood group when needed.*
*Our Objectives include the identification of suitable (compatible) blood donors for patients with specific blood groups; the determination of available and close-by donors for emergency purposes; the ranking of potential donors by proximity to the patient's hospital, past donation history; and the reduction of the amount of time it takes to locate a donor using Data Structures such as Hash Map, Min-Heap and Queue.*
*First, we'll begin by developing the compatibility rules for blood. We'll start this portion of the project with the design and creation of the necessary classes (Donors, Blood Bank, etc.) and also the Match Request Class. This will be followed by an implementation of a Donor registration system. Next, we will implement a hash map in order to organize all the registered donors into categories based upon their blood type. Following this, we will find out which donors qualify. Based on a min heap we will find the donor(s) ranked highest. We will implement a queue next, so that we can handle equal qualified donors, process them and check whether our implementation was correct. Finally and most importantly, we will put it all together and test the entire system using a variety of conditions such as: different blood types of the donors, error checking, etc., so that our system works as efficiently as possible.*

## Project Approach

*It is planned to design the* ***Emergency Blood Donor Matching System*** *through programming and the usage of appropriate data structures. The main idea of the project lies in optimization of the process of choosing the right donor in the case of an emergency situation.*
*Hash Map (Unordered Map) is being implemented to categorize all donors by blood type. Hash Maps can help make the system much more efficient by allowing the program to skip checking every single registered donor and instead find only the ones that have a matching blood type. A min heap or priority queue will be utilized to place all eligible donors in an order based upon how far they live from the hospital and how long it has been since their last donation. The queue would be utilized to select which donor to check next if multiple donors had equal eligibility, utilizing first-in-first-out logic (FIFO).*
*Additionally, when evaluating donor eligibility the time frame since the last donation date will also be taken into consideration. Finally, should no eligible donors exist in close proximity to the hospital; the program will expand its search area.*

## System Architecture

### 1. User Interface Layer ( Web )

- **Users**: Donor, Hospital, Admin
- **Interfaces**: Web
- **Output**: Messages, Notifications, Search Results

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### 2. Request & Matching Layer

- **Request Manager**: Create and manage blood requests
- **Compatibility Checker**: Check blood group compatibility
- **Eligibility Filter**: Filter donors by eligibility (Last donation and Age)

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### 3. Search & Ranking Layer

- **Donor Registry**: Hash Map: Blood type → List of donors
- **Distance Calculator**: Calculate distance from hospital to donors
- **Ranking Engine**: Min-Heap: Rank by distance & score (eligibility priority)
- **Contact Queue**: Queue: Manage equally suitable donors fairly

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### 4. Donor & Hospital Management Layer

- **Donor Manager**: Register, update and manage donors
- **Hospital Manager**: Register and manage hospitals
- **Availability Manager**: Track donor availability status
- **Notification Service**: Send alerts, SMS, push notifications

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### 5. External Services Layer

- **Maps / Geocoding API**: Get coordinates and calculate distance
- **Communication Service**: In-Website Request
- **Authentication Service**: Login, roles and access control

## Project Outcome / Deliverables

*The purpose of our project is to develop an **Emergency Blood Donor Matching System** that will aid individuals in locating potential donors if they require emergency blood donations.*
*Using this **Emergency Blood Donor Matching System**, users (i.e., those requiring blood) will have access to a search tool based on their specific needs (blood type, geographic area, etc.), which will then allow them to find possible donors from all over the world.*
*This project will include an easy-to-use interface for both the patient/relative seeking blood and the blood donor. This will provide essential data regarding the donor's eligibility for donating and whether or not he/she is available at present.*
*In short, the primary outcome of this project will be an emergency blood donor matching system that saves time for both parties involved (donors and recipients).*
*Ultimately, through this project we will demonstrate how programming and technology can solve many everyday problems.*

## Assumptions

*We have made a lot of assumptions in this project. The first set of assumptions that were used in our project was based upon the information given by the donors. This includes the belief that everything they told us about their blood type (A+, A-, etc.), where they lived, what there phone number was, and when they donated blood, etc. was all accurate and true. It has been assumed in our program that the donor data base will be constantly update so as to ensure that the data available to us regarding each potential donor's eligibility is current. Another assumption is made in the process of matching the donor with the patient, which is that the blood compatability law rules will be followed. An additional assumption has been made that if you donate blood, you will wait long enough prior to donating again to meet your requirements. For now, we are going to reduce the complexity associated with determining a person's location so we can determine how close the donor is to the hospital.*

## References

*1. **e-RaktKosh, Government of India** – Blood Availability information, Blood Banks and Blood Donation information.*
*2. **World Health Organization (WHO)** - Blood Safety, Blood Donation and Timely Availability of Blood information.*
*3. **National Blood Transfusion Council (NBTC), Government of India** - Blood Donation and Transfusion Services information.*
*4. **GeeksforGeeks** - Reference Material for Data Structures and Algorithms which can be used for the Donor Matching System.*